1. Identification of Main LKA Components (Part A)

The Lane-Keeping Assist (LKA) system is architected as a discrete-time, closed-loop control system. To ensure dynamic stability and real-time responsiveness, the hardware and software components interact on a strict 0.1-second sampling loop:

- **Perception Sensor (Velodyne VLP-16 LiDAR):** Functions as the primary environmental feedback mechanism. It scans the roadway at 10 Hz, calculating the vehicle's actual lateral position (y) and heading angle (psi) relative to physical lane boundaries (such as curbs or barriers) every 0.1 seconds.
- **Decision Unit (Model Predictive Controller - MPC):** Acts as the computational brain. It receives the positional error data, calculates a 1-second prediction horizon (10 discrete steps), minimizes a defined numerical cost function, and outputs the optimal steering command.
- **Execution Actuator (Toyota Electric Power Steering - EPS):** Physically executes the optimal steering angle (delta) commanded by the MPC, subject to the mechanical limits of the vehicle's steering rack.
- **Plant Model (Kinematic Bicycle Model):** The mathematical representation of the Toyota Corolla Cross's physical geometry, dictating how the steering angle and velocity translate into real-world displacement.

2. Definition of Relevant Physical Parameters (Part B)

To derive a numerically accurate mathematical model ("Digital Twin") for simulation, the following parameters are defined based on the specifications of the Toyota Corolla Cross 1.8V HEV and standard conditions on Pakistani motorways (e.g., M-2). All units are standardized to the SI metric system.

Parameter	Symbol	Value	Unit	Source / Context
Operating Velocity	v	**27.78**	m/s	100 km/h (Standard motorway cruising speed)
Wheelbase	L	**2.64**	m	Toyota Corolla Cross official specification
Vehicle Mass	M	**1525**	kg	Curb weight of the 1.8V HEV variant
Sampling Time	T	**0.1**	s	MPC calculation interval frequency

Parameter	Symbol	Value	Unit	Source / Context
Gravity	G	**9.81**	m/s²	Standard Earth gravity
Road Banking Angle	theta	**1.72**	deg	Approximate 3\% slope on highway curves
Crosswind Force	F-(wind)	**50**	N	Moderate lateral wind disturbance

3. Formulation of Steering Control Equations (Part C)

Using the Kinematic Bicycle Model, we determine the continuous-time rates of change for the vehicle position and orientation by substituting the established physical constants.

3.1 Continuous Yaw Rate (Rotation)

The fundamental equation governing the change in the vehicle's heading is:

$$\dot{\psi} = \frac{v}{L} \tan(\delta)$$

Substituting the specific vehicle parameters:

$$\dot{\psi} = \frac{27.78}{2.64} \tan(\delta)$$

$$\dot{\psi} = \mathbf{10.52} \tan(\delta)$$

Engineering Significance: At a cruising speed of 100 km/h, the Toyota Corolla Cross alters its heading at a rate of 10.52 times the tangent of the commanded steering angle (measured in radians per second).

3.2 Continuous Lateral Velocity (Drift)

The unperturbed lateral drift equation, based solely on the vehicle's heading relative to the lane, is:

$$\dot{y} = v \sin(\psi)$$

$$\dot{y} = \mathbf{27.78} \sin(\psi)$$

3.3 MPC Cost Function Formulation

The objective of the MPC is to find a steering sequence that minimizes the Cost Function (J). Using an empirically tuned weighting strategy to prioritize safety and passenger comfort (10 for lane centering, 5 for heading alignment, 1 for steering smoothness), the numerical optimization problem is defined as:

$$\text{Minimize } J = \sum_{i=1}^{10} \left[10 \cdot (e_{lat,i})^2 + 5 \cdot (e_{\psi,i})^2 + 1 \cdot (\Delta\delta_i)^2 \right]$$

Subject to Toyota Corolla Cross Mechanical Constraints:

Absolute Steering Limits: -35deg <= +35deg

Steering Rate Limit: $|\Delta\delta| \leq 5^\circ$ per 0.1s step (to comply with ISO 11270 safety standards and prevent abrupt manoeuvres).

4. Integration of External Factors and Discretization (Part D)

To ensure the simulation accurately reflects the environmental challenges of the Pakistani market, external disturbances are mathematically integrated, and the continuous model is discretised for digital computation.

4.1 Integrating Environmental Disturbances

Road banking induces a gravitational pull, while crosswinds exert force on the vehicle's side profile. We calculate this combined external lateral acceleration (a) using Newton's Second Law:

$$a_{ext} = g \sin(\theta) + \frac{F_{wind}}{m}$$

Substituting the environmental parameters:

$$a_{ext} = 9.81 \sin(1.72^\circ) + \frac{50}{1525}$$

$$a_{ext} = 0.294 + 0.0327$$

$$\mathbf{a_{ext} = 0.3267 \ m/s^2}$$

Integrating this disturbance into the continuous lateral velocity equation yields:

$$\dot{\mathbf{y}} = \mathbf{27.78} \sin(\psi) + \mathbf{0.3267}$$

4.2 Discretization (The Final Computational Model)

The MPC requires a discrete-time model (utilizing the Euler Method) to predict the vehicle's state 0.1 seconds into the future (k+1). This is achieved by multiplying the continuous rates by the sampling time (T = 0.1).

A. Next Lateral Position (y):

$$y(k+1) = y(k) + T \cdot [\dot{y}]$$

$$y(k+1) = y(k) + 0.1 \cdot [27.78 \sin(\psi(k)) + 0.3267]$$

$$\mathbf{y(k+1) = y(k) + 2.778 \sin(\psi(k)) + 0.03267}$$

B. Next Heading Angle (ψ):

$$\psi(k+1) = \psi(k) + T \cdot [\dot{\psi}]$$

$$\psi(k+1) = \psi(k) + 0.1 \cdot [10.52 \tan(\delta(k))]$$

$$\psi(\mathbf{k+1}) = \psi(\mathbf{k}) + \mathbf{1.052} \tan(\delta(\mathbf{k}))$$

4.3 Sensor Noise and Initial Verification Conditions

To finalize the simulation environment parameters:

- **LiDAR Noise Injection:** The Velodyne VLP-16 has a documented accuracy variance of pm 3 cm. The sensory feedback provided to the MPC is modelled to include Gaussian white noise:

$$y_{meas} = y_{actual} + \mathcal{N}(0, 0.03^2)$$

Initial Verification Conditions: To test the model's recovery capability in Module 3, the simulation will commence at t=0 under a simulated drift scenario:

Initial Lateral Error: y(0) = 0.6 m

Initial Heading Error: psi(0) = 3deg(0.052 rad)

Conclusion

By integrating the exact hardware specifications of the Toyota Corolla Cross and the target environmental parameters into the kinematic framework, a fully quantified, discrete-time mathematical model has been derived. The MPC algorithm will utilize the derived difference equations, y(k+1) and psi(k+1), to predict the vehicle's trajectory and compute the optimal steering sequence (delta) required to counteract the calculated 0.3267 m/s^2 environmental disturbance, ensuring safe lane-keeping operations.